

# Introduction to Natural Language Processing (NLP)

Structured study notes — concepts, definitions, workflow

## 1. What is NLP and Its Applications

### Definition: Natural Language Processing (NLP)

A subfield of Artificial Intelligence (AI) and Linguistics concerned with enabling computers to read, understand, interpret, and generate human language (text or speech) in a way that is both meaningful and useful.

Human language is unstructured, ambiguous, and context-dependent — the same sentence can mean different things depending on tone, culture, or situation. NLP bridges this gap between unstructured human communication and structured data that machines can process using statistics, linear algebra, and machine learning.

### Why NLP is Hard (Core Challenges)

- **Ambiguity:** Words and sentences can have multiple meanings depending on context (e.g., "bank" — river bank vs. financial bank).
- **Figurative language:** Idioms, sarcasm, and slang don't follow literal grammar rules (e.g., "break a leg").
- **Language variability:** Grammar, word order, and meaning differ drastically across languages.
- **Evolving vocabulary:** Words evolve; spelling, abbreviations, and emojis constantly appear ("gr8", "lol", "btw").
- **Context dependency:** Understanding a sentence often needs external/common-sense knowledge not present in the text itself.

### Two Broad Capabilities of NLP

1. **NLU** — Understanding what text or speech means (Natural Language Understanding).
2. **NLG** — Producing coherent, human-like text or speech as output (Natural Language Generation).

### Real-World Applications of NLP

| Application                   | What it does                                      | Example                                        |
|-------------------------------|---------------------------------------------------|------------------------------------------------|
| Machine Translation           | Converts text from one language to another        | Google Translate                               |
| Sentiment Analysis            | Detects opinion/emotion in text                   | Analyzing product reviews as positive/negative |
| Chatbots & Virtual Assistants | Understands queries and responds conversationally | Siri, Alexa, customer-support bots             |
| Text Summarization            | Condenses long text into key points               | News article summarizers                       |

|                                 |                                                |                                              |
|---------------------------------|------------------------------------------------|----------------------------------------------|
| Named Entity Recognition (NER)  | Identifies names, places, organizations, dates | Extracting company names from financial news |
| Spam / Content Filtering        | Classifies text into categories                | Email spam detection                         |
| Speech Recognition              | Converts spoken language to text               | Voice-to-text dictation                      |
| Information Retrieval / Search  | Finds relevant documents for a query           | Search engines                               |
| Text-to-Speech / Speech-to-Text | Converts between spoken and written form       | Screen readers, voice assistants             |
| Question Answering              | Extracts precise answers from text             | IBM Watson, AI search assistants             |

## 2. Overview of NLP Tasks

NLP is not one single task — it is an umbrella term for many distinct problems, each with its own techniques and evaluation methods. Below are the four core tasks introduced in this session.

### 2.1 Text Classification

#### Definition: Text Classification

The task of assigning a predefined category or label to a given piece of text, based on its content.

#### How it Works

- Step 1** — A labeled dataset is collected — each text sample has a known category (e.g., "spam" / "not spam").
- Step 2** — Raw text is cleaned and converted into numerical features (see Section 3 — Feature Extraction).
- Step 3** — A machine learning model (Naive Bayes, Logistic Regression, SVM) or deep learning model (LSTM, Transformer) is trained on these features.
- Step 4** — The trained model predicts the category of new, unseen text.

#### Common Examples

- Spam detection (spam vs. ham)
- Topic categorization (sports, politics, tech)
- Language detection
- Intent classification in chatbots

### 2.2 Sentiment Analysis

#### Definition: Sentiment Analysis (Opinion Mining)

A specialized type of text classification that determines the emotional tone or opinion expressed in text — typically positive, negative, or neutral.

## Approaches

| Approach               | Description                                                                                     | Example Tool/Technique             |
|------------------------|-------------------------------------------------------------------------------------------------|------------------------------------|
| Lexicon-based          | Uses a dictionary of words pre-scored with sentiment polarity; sums/averages scores in the text | VADER, SentiWordNet                |
| Machine Learning-based | Trains a classifier on labeled sentiment data using extracted features                          | Naive Bayes, SVM on TF-IDF vectors |
| Deep Learning-based    | Uses neural networks to learn contextual sentiment patterns                                     | LSTM, BERT fine-tuning             |

## Levels of Sentiment Analysis

- **Document-level:** Overall sentiment of the entire document.
- **Sentence-level:** Sentiment of each sentence separately.
- **Aspect-based:** Sentiment about a specific feature/aspect (e.g., "battery life is bad but camera is great").

## 2.3 Named Entity Recognition (NER)

### Definition: Named Entity Recognition (NER)

The task of locating and classifying "named entities" in text into predefined categories such as person names, organizations, locations, dates, and monetary values.

## How it Works

7. **Step 1** — Text is split into tokens (words).
8. **Step 2** — Each token is analyzed using its context (surrounding words), capitalization, and patterns.
9. **Step 3** — A sequence labeling model (CRF, BiLSTM-CRF, or Transformer-based like spaCy/BERT-NER) tags each token with an entity type (or "O" for none) using a scheme like BIO tagging (Begin, Inside, Outside).
10. **Step 4** — Consecutive tokens with the same entity tag are merged into a single entity span.

## Common Entity Types

| Tag       | Meaning                           | Example                    |
|-----------|-----------------------------------|----------------------------|
| PERSON    | Names of people                   | "Elon Musk"                |
| ORG       | Organizations/companies           | "Google", "United Nations" |
| GPE / LOC | Geopolitical entities / locations | "India", "Paris"           |
| DATE      | Dates and time expressions        | "July 8, 2026"             |
| MONEY     | Monetary values                   | "\$500"                    |

## 2.4 Machine Translation

### Definition: Machine Translation (MT)

The task of automatically converting text or speech from a source language into a target language while preserving meaning.

### Evolution of Translation Techniques

11. **Rule-Based MT (RBMT)** — Word-for-word substitution using bilingual dictionaries and hand-written grammar rules. Rigid and inaccurate for complex sentences.
12. **Statistical MT (SMT)** — Learns translation patterns (phrase-to-phrase mappings) from large bilingual corpora using probability models.
13. **Neural MT (NMT)** — Uses an Encoder–Decoder neural network (often Transformer-based) that reads the entire source sentence and generates the target sentence, capturing long-range context far better than SMT.

*Modern systems like Google Translate use NMT with attention mechanisms/Transformers to achieve fluent, context-aware translations.*

## 3. The NLP Workflow

Every NLP project — regardless of the specific task — follows a common end-to-end pipeline. Understanding this pipeline is essential before diving into any individual task.

### Pipeline Overview

**Text Collection → Preprocessing → Feature Extraction → Modeling**

### 3.1 Text Collection

#### Definition: Text Collection

The process of gathering raw textual data from various sources to build the dataset that the NLP system will learn from or operate on.

#### Common Sources

- Web scraping (news sites, blogs, forums)
- APIs (Twitter/X API, Reddit API, Google Places API)
- Public datasets (Kaggle, IMDB reviews, UCI repository, Hugging Face Datasets)
- Internal company data (support tickets, chat logs, emails)
- Surveys, PDFs, scanned documents (via OCR)

## Key Considerations

- **Volume:** Enough samples per class/category to train a reliable model.
- **Diversity:** Balanced representation of all classes/topics to avoid bias.
- **Quality:** Correct labels (for supervised tasks) — mislabeled data directly hurts model performance.
- **Legal/ethical:** Handling personal or sensitive data responsibly (GDPR, anonymization).

## 3.2 Text Preprocessing

### Definition: Text Preprocessing

The set of steps used to clean and standardize raw text, transforming it from messy, human-written language into a consistent form suitable for feature extraction.

Raw text is "dirty" — it contains punctuation, inconsistent casing, stopwords, typos, and irrelevant symbols (URLs, HTML tags, emojis). Preprocessing reduces noise and vocabulary size, which improves model accuracy and training efficiency.

### Standard Preprocessing Steps (in typical order)

14. **Cleaning** — Remove HTML tags, URLs, special characters, extra whitespace, and (optionally) numbers.
15. **Lowercasing** — Convert all text to a single case (usually lowercase) so "Cat" and "cat" are treated as the same token.
16. **Tokenization** — Split text into smaller units — words (word tokenization) or sentences (sentence tokenization).
17. **Stopword Removal** — Remove common words that carry little meaning ("the", "is", "a", "and") to focus on content-bearing words.
18. **Punctuation Removal** — Remove punctuation marks (periods, commas, quotes) that don't usually add semantic value for many tasks.
19. **Stemming / Lemmatization** — Reduce words to their root/base form.
20. **Spelling Correction** — Correct common misspellings (optional, dataset-dependent).

### Stemming vs. Lemmatization

| Aspect     | Stemming                                       | Lemmatization                                                                     |
|------------|------------------------------------------------|-----------------------------------------------------------------------------------|
| Definition | Chops word endings using crude heuristic rules | Reduces word to its dictionary base form (lemma) using vocabulary + grammar rules |
| Output     | May not be a real word                         | Always a valid word                                                               |
| Speed      | Faster                                         | Slower (needs POS info + lookup)                                                  |
| Example    | "studies" → "studi"                            | "studies" → "study"                                                               |

|           |                                  |                           |
|-----------|----------------------------------|---------------------------|
| Example 2 | "better" → "better"              | "better" → "good"         |
| Tools     | Porter Stemmer, Snowball Stemmer | WordNet Lemmatizer, spaCy |

### 3.3 Feature Extraction

#### Definition: Feature Extraction

The process of converting cleaned, tokenized text into numerical vectors that machine learning algorithms can process, since models only understand numbers, not raw words.

#### Common Techniques

| Technique             | Idea                                                                           | Limitation                                                                     |
|-----------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Bag of Words (BoW)    | Represents text as word counts, ignoring order/grammar                         | Ignores word order & context; large sparse vectors                             |
| TF-IDF                | Weights words by frequency in a document vs. rarity across all documents       | Still ignores word order & semantic meaning                                    |
| N-grams               | Captures sequences of N consecutive words (e.g., bigrams) to retain some order | Vocabulary size grows quickly                                                  |
| Word Embeddings       | Dense vectors that capture semantic meaning; similar words → similar vectors   | Needs large corpora; fixed vector per word (no context) for classic embeddings |
| Contextual Embeddings | Vector for a word depends on surrounding context (e.g., BERT)                  | Computationally expensive                                                      |

#### TF-IDF Formula (most common baseline technique)

$$\begin{aligned} \text{TF}(t, d) &= (\text{Number of times term } t \text{ appears in document } d) / (\text{Total terms in } d) \\ \text{IDF}(t) &= \log(\text{Total number of documents} / \text{Number of documents containing } t) \\ \text{TF-IDF}(t, d) &= \text{TF}(t, d) \times \text{IDF}(t) \end{aligned}$$

A high TF-IDF score means the term is frequent in a specific document but rare across the whole corpus — making it a strong distinguishing feature for that document.

### 3.4 Modeling

#### Definition: Modeling

The stage where a machine learning or deep learning algorithm is trained on the numerical features to learn patterns and make predictions (classification, generation, etc.).

#### Typical Modeling Steps

- Step 1** — Divide data into training, validation, and test sets.

22. **Step 2 — Choose** — Choose an appropriate algorithm based on task, data size, and interpretability needs.
23. **Step 3 — Fit** — Fit the model on the training features and labels.
24. **Step 4 — Evaluate** — Use accuracy, precision, recall, F1-score (classification) or BLEU score (translation), etc.
25. **Step 5 — Tune** — Adjust hyperparameters, try different features/models to improve results.
26. **Step 6 — Deploy** — Save the model and use it to make predictions on new, real-world text.

### Model Families Used in NLP

| Category                        | Examples                                             | Best Suited For                                                 |
|---------------------------------|------------------------------------------------------|-----------------------------------------------------------------|
| Classical ML                    | Naive Bayes, Logistic Regression, SVM, Random Forest | Small-medium datasets, fast baselines                           |
| Sequence Models (Deep Learning) | RNN, LSTM, GRU                                       | Sequential/contextual patterns, moderate data                   |
| Transformer-based               | BERT, GPT, RoBERTa, T5                               | State-of-the-art accuracy, large data, contextual understanding |

## 4. Quick-Recall Summary

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- **NLP:** NLP enables machines to understand and generate human language; used in translation, sentiment analysis, chatbots, search, and more.
- **Text Classification:** Assigns predefined labels to text (e.g., spam detection, topic tagging).
- **Sentiment Analysis:** Detects positive/negative/neutral opinion in text; can be lexicon-based, ML-based, or deep learning-based.
- **NER:** Identifies and classifies names, organizations, locations, dates, etc. in text.
- **Translation:** Converts text between languages; evolved from rule-based → statistical → neural (Transformer) approaches.
- **NLP Workflow:** Text Collection → Preprocessing (clean, tokenize, remove stopwords, stem/lemmatize) → Feature Extraction (BoW, TF-IDF, embeddings) → Modeling (train, evaluate, tune, deploy).